ISAI MELENDEZ

Houston, Texas | isaimel001@gmail.com | LinkedIn | Website

Education

Yale University — New Haven, CT
Joint Major: Computing and the Arts (B.A.) | Concentration: Art | GPA 3.81/4.0 — *2022 – 2026*
Relevant Courses: Interactive Design and the Internet, Data Structures and Programming Techniques, Object-Oriented Programming, Building Game Engines

St. John's School — Houston, Texas
GPA 3.7/4.0 | SAT 1520 — *2018 – 2022*

Work Experience

Undergraduate Tour Guide — New Haven, CT
Yale University Art Gallery — *Sep 2023 – May 2026*
- Led weekly 60-minute tours for groups of 10-20 visitors, improvising and modifying to audience questions and engagement.
- Collaborated with the 20-person Gallery Guide team and public education staff, providing and adopting feedback on tours.
- Adapted in-person tours and self-written reports into concise 400-word scripts and recorded 4-minute audio tours, accessible on the Smartify app, a global museum content platform.

Graphic Designer — New Haven, CT
Jonathan Edwards (JE) College — *Sep 2022 – May 2026*
- Upheld JE College's identity across merchandise for a student body of 500, adhering to established brand guidelines.
- Designed merchandise for three consecutive years of the annual Tulip Princess tradition, featuring each titleholder's portrait.
- Produced 50+ promotional flyers, videos, and newsletters in Adobe Suite for college events, traditions, and updates.
- Created 30+ graphics and reusable templates for recurring use.
- Collaborated with college staff and peers to workshop designs, incorporating suggestions and critiques.

Design, Communications, and Data Summer Intern — Houston, TX
St. John's High School (SJS) — *May 2022 – July 2024*
- Created infographics, videos, and other promotional material to advertise the SJS Phonathon, a fundraising event raising an average of $300k annually.
- Interviewed notable alumni across diverse professional fields, tailoring questions to highlight their career histories, motivations, and future aspirations; authored articles published in the SJS Alumni Magazine.
- Contributed to the transition of donation records from a physical to a digital database. Consolidated alumni board members and membership records from disorganized sources.

Leadership

Jonathan Edwards College Council — New Haven, CT
Vice President/President — *Aug 2024 – May 2025*
- Led a council of 20+ members in brainstorming and executing college-wide events by managing budgets, coordinating with staff, and delegating student volunteers.
- Oversaw a monthly budget of $2,500, recording event attendance and spending data to maximize savings.
- Maintained weekly communication and collaboration with the Head of College, Dean of College, and Assistant Director of Operations to finalize event proposals and logistics.

Projects

Preserving Artistry in Automated Terrain Generation — *Aug 2025 – May 2026*
- Developed a 3D terrain generator that gradually fills grid cells with blocks derived from a supplied modular asset set.
- Automated adjacency detection between pairs of blocks in Python/Blender, compiling adjacency constraints for each block.
- Implemented the Wave Function Collapse algorithm in C#/Unity, ensuring a seamlessly assembled output by propagating the state of a newly defined grid cell outwards and removing adjacency-breaking possibilities from undefined cells.

BAPI Farms — *Aug 2025 – Dec 2025*
- Built a 2D game demo and tilemap editor with three peers using a custom game engine built from scratch in D.
- Implemented player movement and object interaction using the Command and State patterns, action mapping, and acceleration-based movement.

Portfolio Website — *June 2026 – July 2026*
- Produced a personalized Website from scratch with HTML, CSS, and JavaScript, hosted online via GitHub Pages.
- Designed the website to easily add new content through separate data files without altering preexisting code.

Skills & Interests

Languages: English (native), Spanish (proficient), French (conversational)

Programming: C, C#, C++, D, Python, JavaScript, HTML/CSS, LaTeX

Software: Microsoft Office (Excel, Word, PowerPoint), Unity, Blender, Adobe Suite (Photoshop, Premier Pro, Fresco)

Skills: Web Development, Problem Solving, Responsible Use of AI, Project Management, Collaboration, Communication, Research

Interests: Software Development, Video Game Creation, App Development, Art Through Technology, Data Analysis